4) i) 158.58^{-2}

$= (1\times2^2)+(5\times2^1)+(8\times2^0)+(5\times2^{-1})+(8\times2^{-2})$

2) -8 subtract from 23 means addition of 13 and 8

13 — 1101
8 — 1000
10101 (21)

3) 10101.101^{-3} $= 1\times2^4+2^2+2^0+2^{-1}+2^{-3}$

$= 16+4+1+\frac{1}{2}+\frac{1}{8}$

$= 21+\frac{5}{8}$

$= 21\frac{5}{8}$

$= 21.625$

4) 237.5^{-1}

$= (2\times8^2)+(3\times8^1)+(7\times8^0)+(5\times8^{-1})$

5) i) 158.58^{-2}

$= (1\times10^2)+(5\times10^1)+(8\times10^0)+(5\times10^{-1})+(8\times10^{-2})$

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v)

9A8 8

$9 \; 10 \; 8 \; 8$

$= (9 \times 16^2) + (10 \times 16^1) + (8 \times 16^0) + (8 \times 16^{-1})$

vi) add -8 and 12 means substraction of 8 from

12 7 10001

8 → - 1000

01001 (9)

vii) 10011011

Octal → group from right

010 011 011 011

2 3 3 3

2333_8

hexa → 01001101 1011

4D B

viii)

6 5 4 3 2 1 0

1 1 0 0 1 0 0

$= 2^6 + 2^5 + 2^2 + 2^0$

$= 64 + 32 + 4 + 1 = 100$

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viii) 1100 11

$2^5+2^4+2^1+2^0$

$=32+16+2+1$

$= 51$

$=$ 10100001 (BCD)

ix)

10101110 ⟶ 11001011

A	B	
0	0	0
0	1	1
1	0	1
1	1	0

x) 10111001 0

01000111 0

1

01000110

$= -2^8+2^6+2^5+2^4+2^1$

$= -256+64+32+16+2$

$= -142$

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2) a) i) $A + \bar{A}B$

$= A + AB + \bar{A}B$

$= A + B(A + \bar{A})$ $[A + \bar{A} = 1]$

$= A + B \cdot 1$

$= A + B$

ii) $(A+B)(A+C)$

$= A \cdot A + AC + BA + BC$

$= A + AC + AB + BC$

$= A(1 + C + B) + BC$

$= A \cdot 1 + BC$

$= A + BC$

—o—

b)

$$\begin{array}{r} 0111 \\ 1001 \\ \hline 10000 \end{array}$$

7
9
16

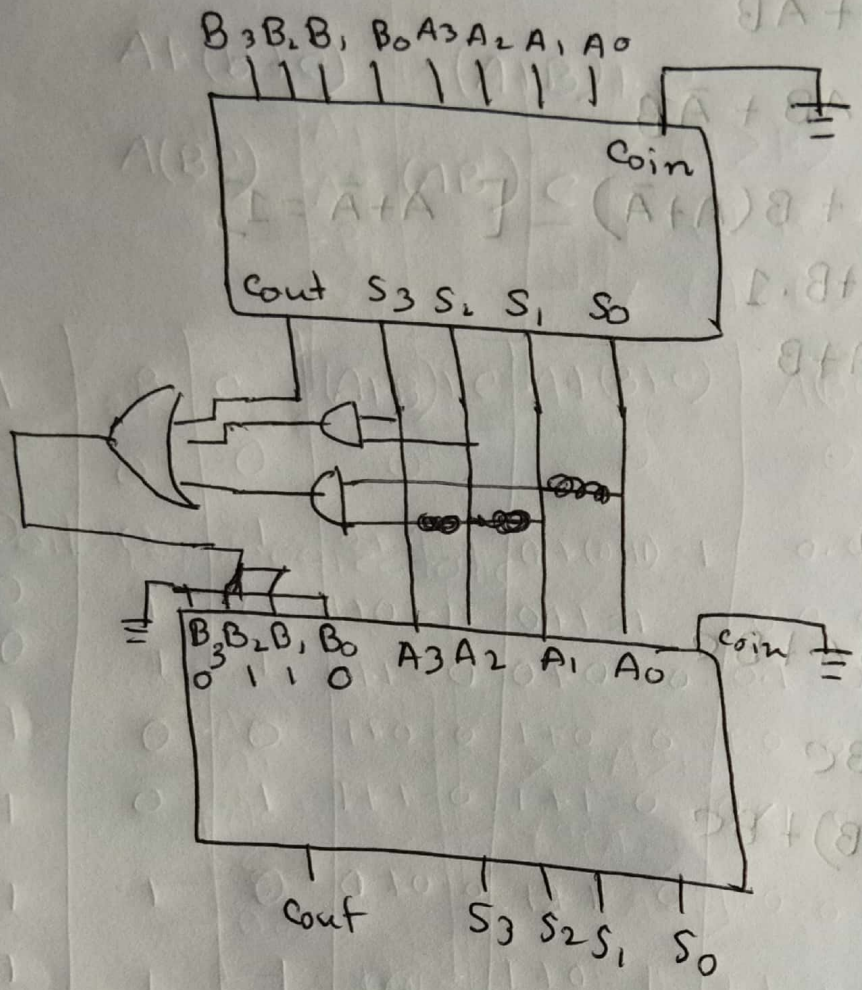
if number is greater than 9 or there is carry from one group (0000) to another we have to add 0110

$$\begin{array}{r} 0111 \\ 1001 \\ \hline 00010000 \\ 0110 \\ \hline 00010110 \end{array}$$

1 6

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Circuit diagram



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c) Fan defines the maximum numbers of digital inputs that a single logic gate can accept

Gray code are not ~~used~~ to represent a numerical quantity. but rather a representation of unique identifiers without numerical significance. So addition of numbers in Gray code creates ~~an~~ incorrect answers.

—o—

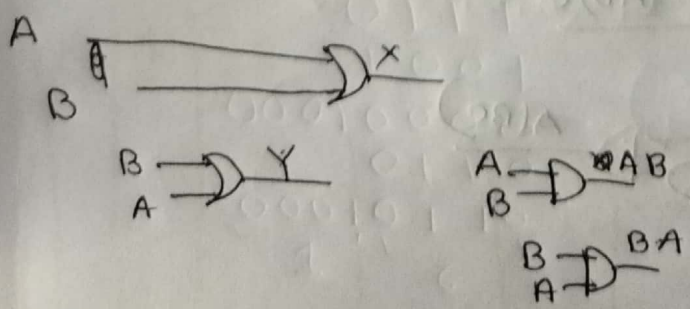
d) Boolean algebra is the branch of algebra in which ~~vare~~ values of variables are the truth values: true and false.

Laws of Boolean algebra

1) Commulative law

$$A+B = B+A$$
$$AB = BA$$

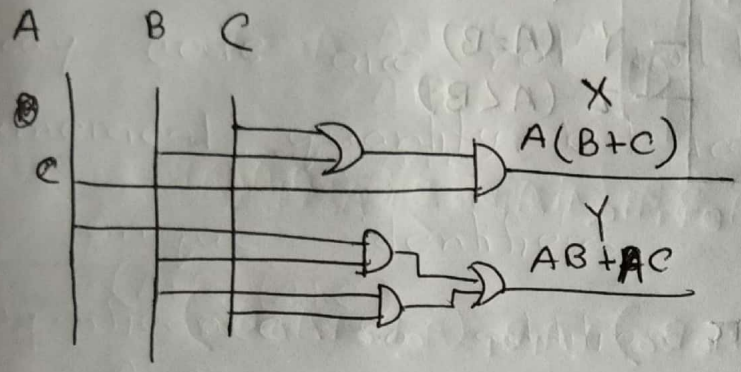
A	B	A+B	B+A	AB	BA
0	1	1	1	0	0
1	0	1	1	0	0
0	0	0	0	0	0
1	1	1	1	1	1



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ii) Distributive

$A(B+C) = AB + AC$



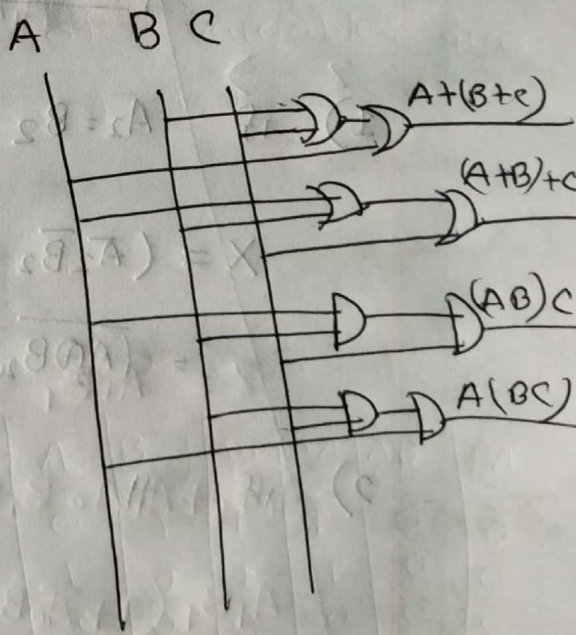
A	B	C	X	Y
0	0	0	0	0
0	0	1	0	0
0	1	0	0	0
0	1	1	0	0
1	0	0	0	0
1	0	1	1	1
1	1	0	1	1
1	1	1	1	1

iii) Associative law

$A + (B+C) = (A+B) + C$

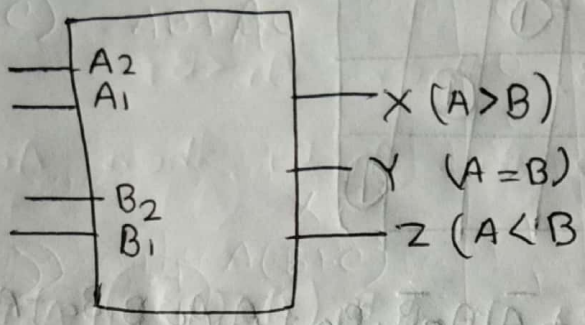
$A(BC) = (AB)C$

A	B	C	A+(B+C)	(A+B)+C	A(BC)	(AB)C
0	0	0	0	0	0	0
0	0	1	1	1	0	0
0	1	0	1	1	0	0
0	1	1	1	1	0	0
1	0	0	1	1	0	0
1	0	1	1	1	0	0
1	1	0	1	1	0	0
1	1	1	1	1	1	1

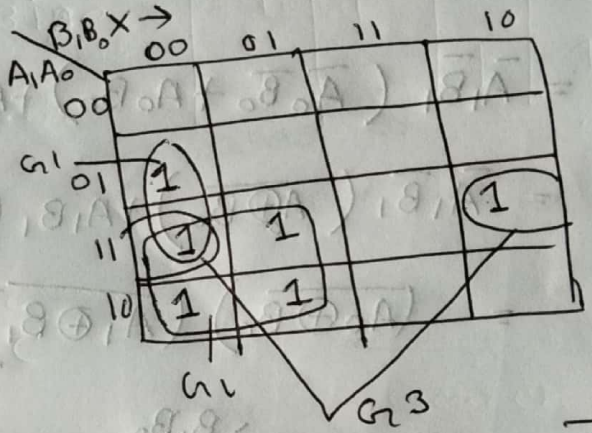


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3) a) 2 bit comparator



A1	A0	B1	B0	X	Y	Z
0	0	0	0	0	1	0
0	0	0	1	0	0	1
0	0	1	0	0	0	1
0	0	1	1	0	0	1
0	1	0	0	1	0	0
0	1	0	1	0	1	0
0	1	1	0	0	0	1
0	1	1	1	0	0	1
1	0	0	0	1	0	0
1	0	0	1	1	0	0
1	0	1	0	0	1	0
1	0	1	1	0	0	1
1	1	0	0	1	0	0
1	1	0	1	1	0	0
1	1	1	0	1	0	0
1	1	1	1	0	1	0



$X = A_0\bar{B_1}\bar{B_0} + A_1\bar{B_1} + A_1A_0\bar{B_0}$

$= A_1\bar{B_1} + A_0\bar{B_1}\bar{B_0}\bar{A_1} + \bar{A_0}B_1\bar{B_0}A_1$

$+ A_1A_0B_1\bar{B_0} + A_1A_0\bar{B_1}\bar{B_0}$

$= \overline{A_1B_1} + \bar{B_1}\bar{B_0}(A_1A_0 + A_1\bar{A_0})$

$= A_1\bar{B_1} + A_0B_1\bar{B_0}(A_1 +$

$= A_1\bar{B_1}\left(1 + A_0\bar{B_0}\right) + A_0\bar{B_0}\right)$

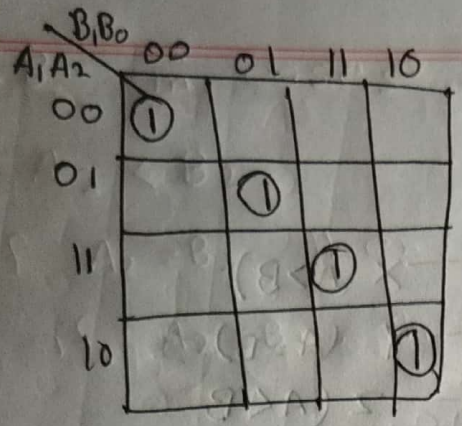
$+ \bar{A_1}A_0\bar{B_1}\bar{B_0} + A_1A_0B_1\bar{B_0}$

$= A_1\bar{B_1} + A_0\bar{B_0}(\bar{A_1}\bar{B_1} + A_1B_1)$

$= A_1\bar{B_1} + A_0\bar{B_0}\left(\overline{A_1 \oplus B_1}\right)$

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Y



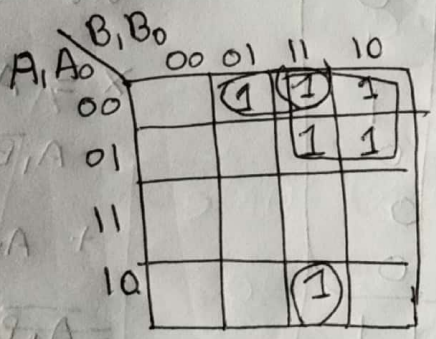
$$Y = \bar{A}_0\bar{A}_1\bar{B}_1\bar{B}_0 + \bar{A}_1A_0\bar{B}_1B_0 + A_1A_0B_1B_0 + A_1\bar{A}_0B_1\bar{B}_0$$

$$= \bar{A}_1\bar{B}_1(\bar{A}_0\bar{B}_0 + A_0B_0) + A_1B_1(A_0B_0 + \bar{A}_0\bar{B}_0)$$

$$= \bar{A}_1\bar{B}_1\overline{(A_0 \oplus B_0)} + A_1B_1\overline{(A_0 \oplus B_0)}$$

$$= \overline{(A_0 \oplus B_0)}\,\overline{(A_1 \oplus B_1)}$$

Z →



$$Z = \bar{A}_1B_1 + \bar{A}_1\bar{A}_0B_0 + \bar{A}_0B_1B_0$$

$$= \bar{A}_1B_1 + \bar{A}_1\bar{A}_0B_1B_0 + \bar{A}_1\bar{A}_0\bar{B}_1B_0 + A_1\bar{A}_0B_1B_0 + A_1\bar{A}_0B_1B_0$$

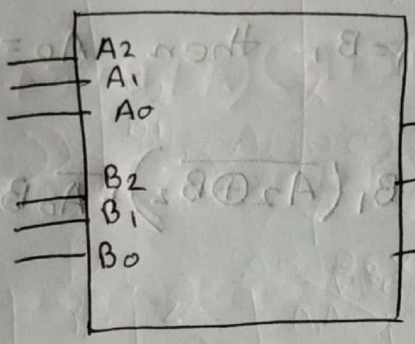
$$= \bar{A}_1B_1(1 + \bar{A}_0B_0) + \bar{A}_0B_0(A_1B_1 + \bar{A}_1\bar{B}_1)$$

$$= \bar{A}_1B_1 + \bar{A}_0B_0\,\overline{(A_1 \oplus B_1)}$$

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3 bit Circuit → তুলনা

3 bit comparator



the output is A>B in cases of

$A_2 > B_2$

$A_2 = B_2$ and $A_1 > B_1$

$A_2 = B_2$, $A_1 = B_1$ and $A_0 > B_0$

$$X = A_2\overline{B_2} + A_1\overline{B_1}(\overline{A_2 \oplus B_2}) + A_0\overline{B_0}(\overline{A_2 \oplus B_2})(\overline{A_1 \oplus B_1})$$

the output is A=B in cases of

$A_0 = B_0$, $A_1 = B_1$ and $A_2 = B_2$

$$Y = (\overline{A_0 \oplus B_0})(\overline{A_1 \oplus B_1})(\overline{A_2 \oplus B_2})$$

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A < B in cases if

$A_2 < B_2$

if $A_2 = B_2$ then $A_1 < B_1$

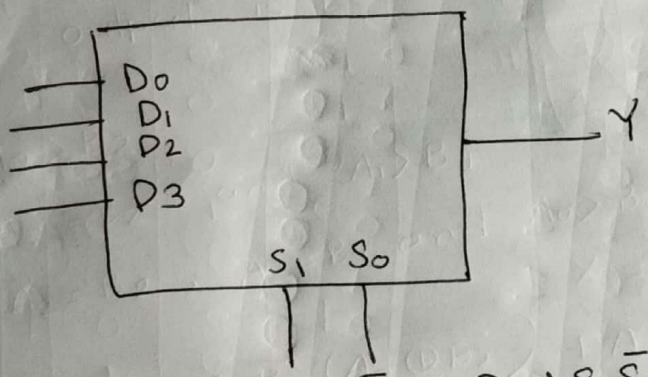
if $A_2 = B_2$, $A_1 = B_1$ then $A_0 = B_0$

$$z = \overline{A}_2 B_2 + \overline{A}_1 B_1 (\overline{A_2 \oplus B_2}) + \overline{A}_0 B_0 (\overline{A_2 \oplus B_2})(A_1 \oplus B_1)$$

—o—

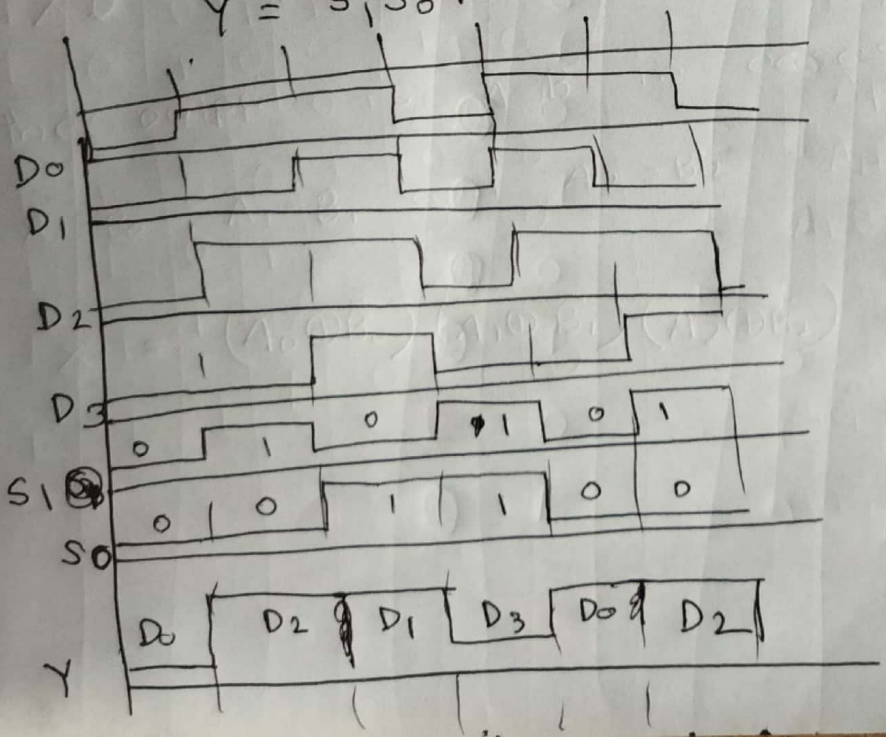
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b) A multiplexer is a device that allows digital information from several sources to be routed into a single line for transmission over that line to a common destination. It has data select input which permit digital data on any one of the inputs to be switched input line.

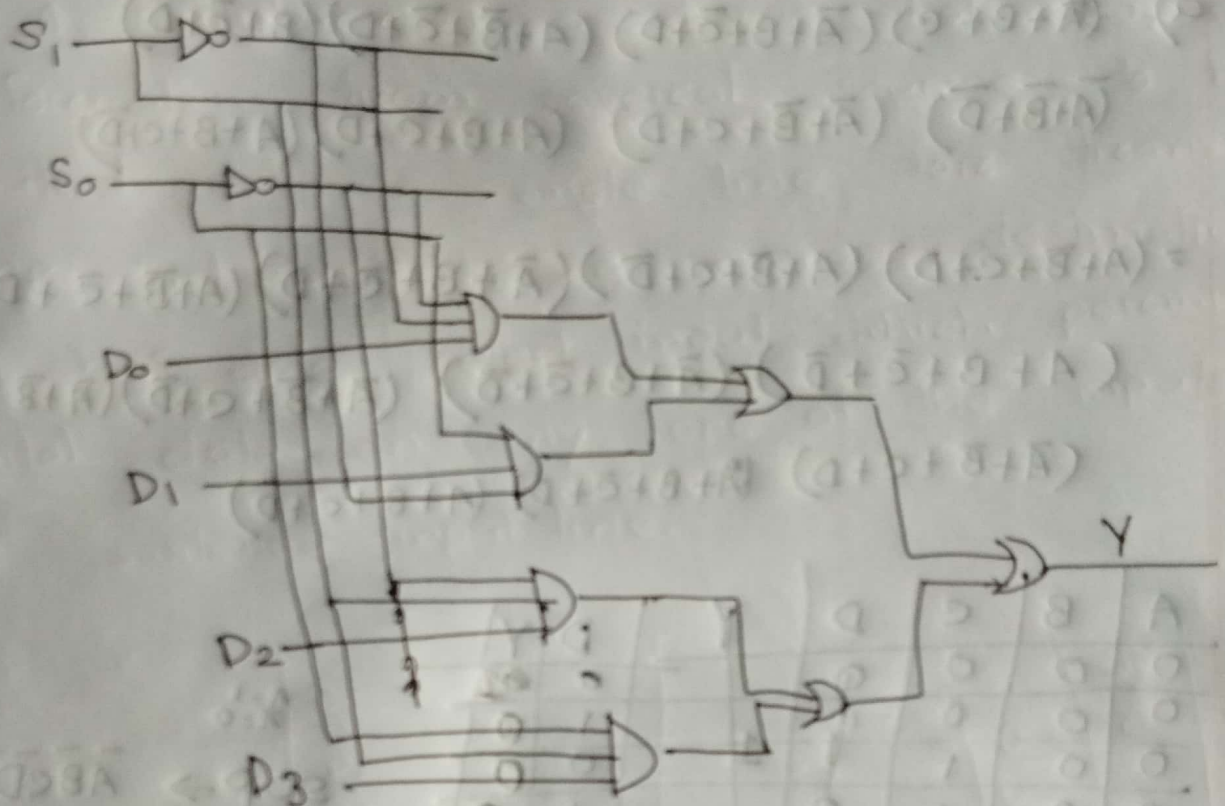


S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_2 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$$



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a) $(\bar{A}+\bar{B}+C)(\bar{A}+B+\bar{C}+D)(A+\bar{B}+\bar{C}+D)(B+\bar{C}+\bar{D})$
$(\bar{A}+\bar{B}+\bar{D})(\bar{A}+\bar{B}+C+D)(A+B+\bar{C}+D)(A+B+C+\bar{D})$

$=(A+\bar{B}+C+D)(A+\bar{B}+C+\bar{D})(\bar{A}+B+\bar{C}+D)(A+\bar{B}+\bar{C}+D)$
$(A+B+\bar{C}+\bar{D})(\bar{A}+B+\bar{C}+\bar{D})(\bar{A}+\bar{B}+C+\bar{D})(\bar{A}+\bar{B}+\bar{C}+\bar{D})$
$(\bar{A}+\bar{B}+C+D)(\bar{A}+B+\bar{C}+D)(A+B+\bar{C}+\bar{D})$

$A=0$
$\bar{A}=1$

A	B	C	D	Y
0	0	0	0	
0	0	0	1	
0	0	1	1	1
0	0	1	0	
0	1	0	1	1
0	1	0	0	1
0	1	1	1	
0	1	1	0	1
1	0	0	1	
1	0	0	0	
1	0	1	1	1
1	0	1	0	1
1	1	0	1	1
1	1	0	0	
1	ϕ	1	1	
1	ϕ	1	0	

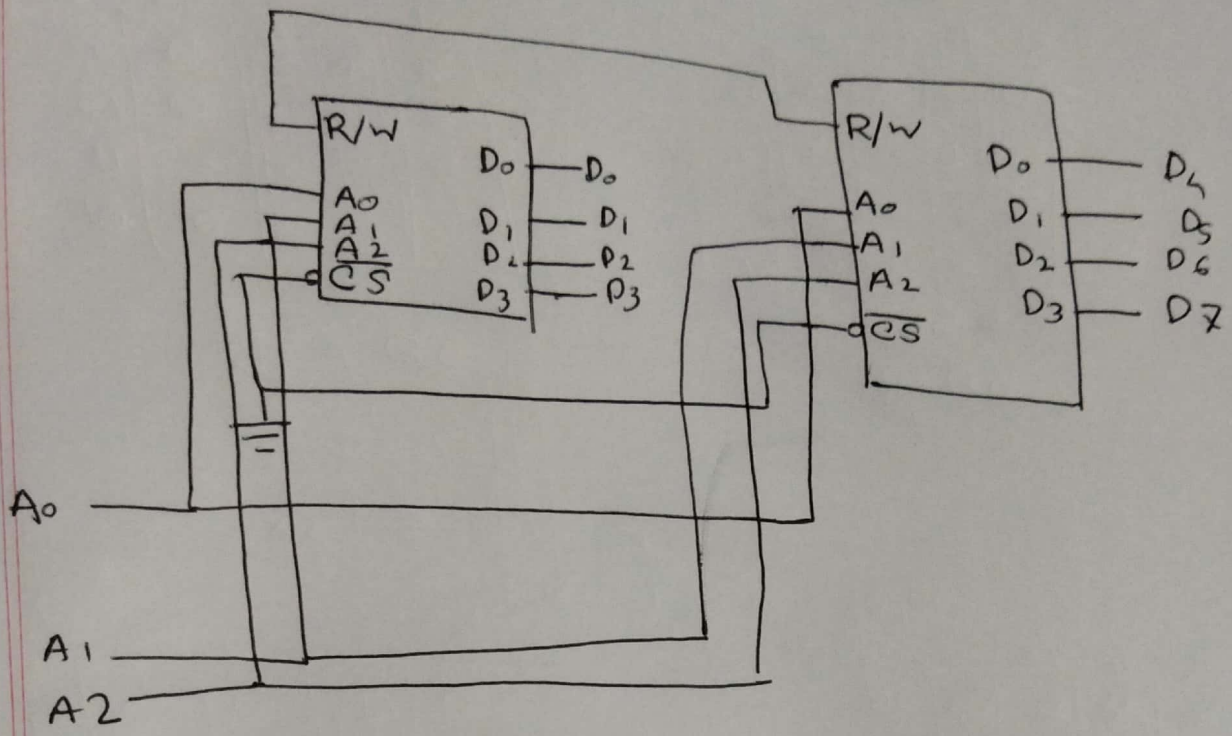
D	Y
0	1
1	0
0	0
1	0
0	0
1	0
0	0
1	1
0	1
1	1
0	0
1	0
0	0
1	0
0	1
1	0

$A=1$
$\bar{A}=0$

SOP → $\bar{A}\bar{B}C\bar{D}+\bar{A}BC\bar{D}+A\bar{B}\bar{C}\bar{D}+A\bar{B}\bar{C}D+\bar{A}\bar{B}C\bar{D}$

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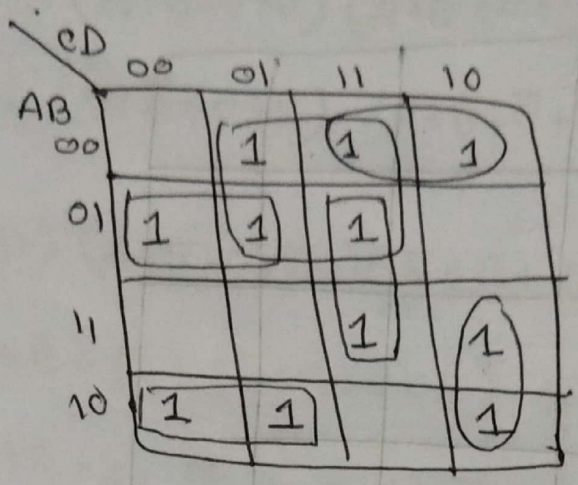
4) b) 16 x 8 RAM using 8x4 RAM



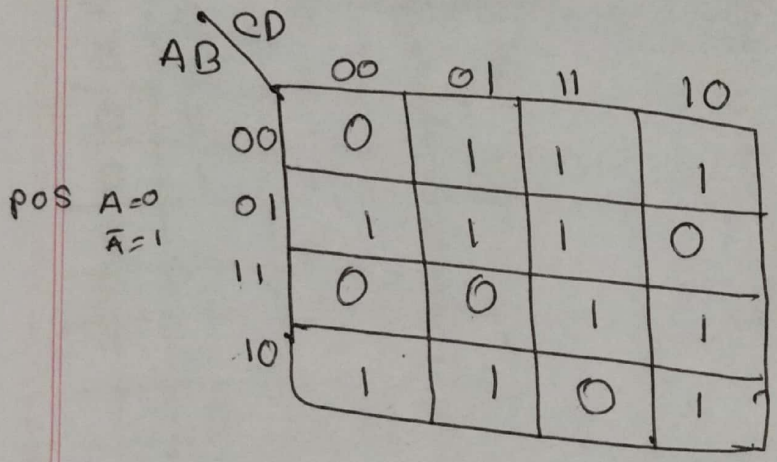
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b)

SOP → 1
POS - 0



SOP → $\bar{A}D + \bar{A}\bar{B}C + \bar{A}B\bar{C} + BCD + ACD + A\bar{B}\bar{C}$



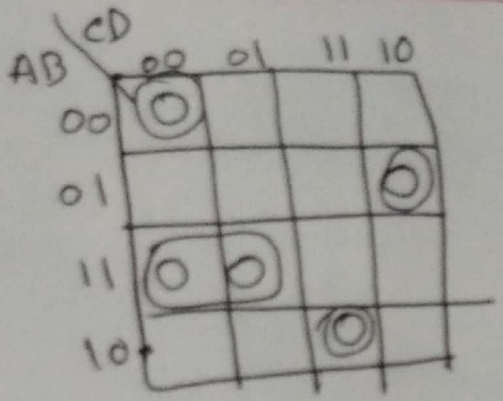
SPOS → $ABCD + A\bar{B}\bar{C}D + \bar{A}\bar{B}CD + A\bar{B}CD + ABCD$

$\bar{A}\bar{B}C\bar{D} + A\bar{B}\bar{C}\bar{D}$

SPOS → $(A+B+C+D)(A+\bar{B}+\bar{C}+D)(\bar{A}+\bar{B}+C+\bar{D})$

$(\bar{A}+\bar{B}+C+\bar{D})(\bar{A}+B+\bar{C}+\bar{D})$

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minimum POS

$= (A+B+C+D)(\bar{A}+\bar{B}+\bar{C})$
$(\bar{A}+\bar{B}+\bar{C}+\bar{D})(\bar{A}+B+\bar{C}+\bar{D})$

— 6 —

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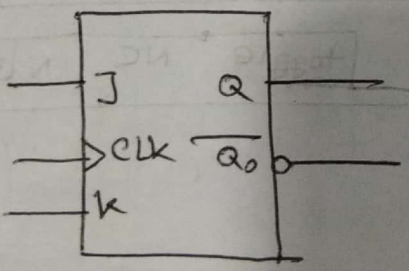
Flip-flop

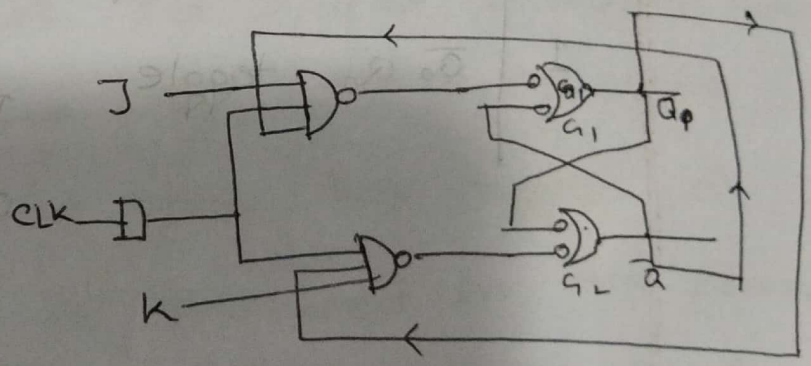
2019 5 a) flip-flops are synchronous bistable device also known as bistable multivibrators. Here synchronous means the output changes state only at a specific point on the triggering input called Clock (CLK)

J-k flipflop -

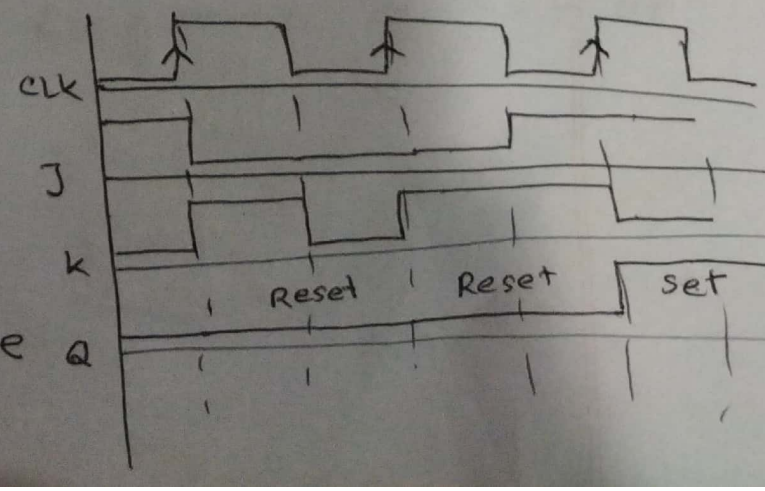
~~the inputs are J and k in J-k flipflop~~

The J and k inputs ~~are~~ ~~trigger~~ of the Jk flip flop are synchronous input because the data on input are transferred to the flip flop's output only on triggering edge of clock pulse.





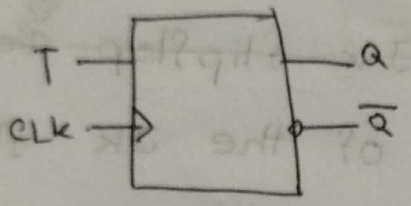
CLK	J	k	Q $\bar{Q}$
↓	0	0	Q_0 $\bar{Q}_0$ NC
↓	0	1	0 1 Reset
↓	1	0	1 0 set
↓	1	1	$\bar{Q}_0$ Q_0 toggle

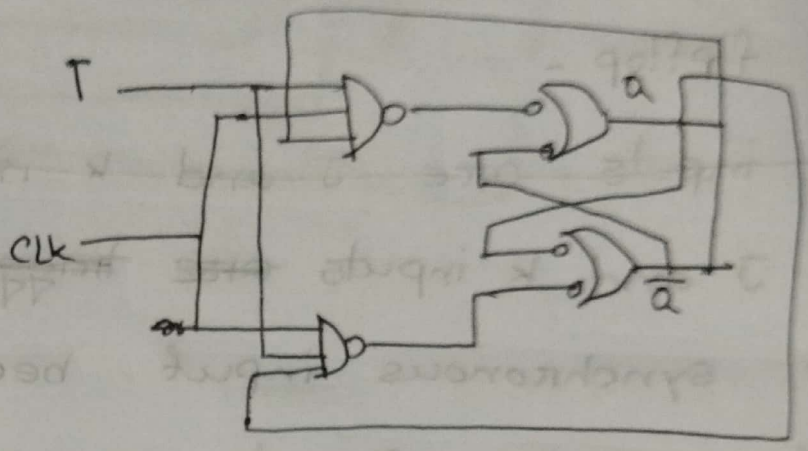


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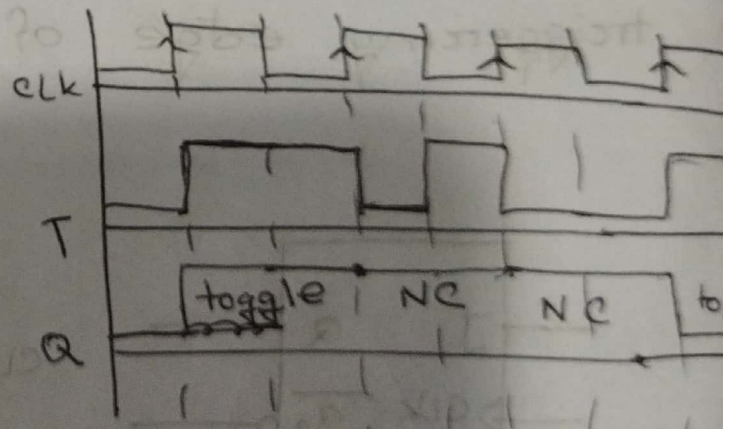
T flip flop

In T flip flop the term T defines toggle. It means the next output stage is changed with the complement of present state output.





T	Q	$\overline{Q}$	
0	Q_0	$\overline{Q_0}$	No change
1	$\overline{Q_0}$	Q_0	toggle



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5) b) $\bar{A}BC\bar{D} + \bar{A}BCD + A\bar{B}CD + AB\bar{C}D + ABC\bar{D} + \bar{A}\bar{B}CD + \bar{A}BCD + \bar{A}BC\bar{D}$

A	B	C	D	Y	
0	0	0	0	0	D_0
0	0	0	1	0	D_1
0	0	1	0	0	D_2
0	0	1	1	1	D_3
0	1	0	0	0	D_4
0	1	0	1	0	D_5
0	1	1	0	1	D_6
0	1	1	1	1	D_7
1	0	0	0	0	D_8
1	0	0	1	0	D_9
1	0	1	0	0	D_{10}
1	0	1	1	1	D_{11}
1	1	0	0	0	D_{12}
1	1	0	1	1	D_{13}
1	1	1	0	1	D_{14}
1	1	1	1	1	D_{15}

4x1 MUX

A, B → select line

C, D → input

	I_0	I_1	I_2	I_3
$\bar{C}\bar{D}$	0	4	8	12
$\bar{C}D$	1	5	9	13
$C\bar{D}$	2	6	10	14
CD	3	7	11	15
	CD	C	$C\bar{D}$	$C+D$

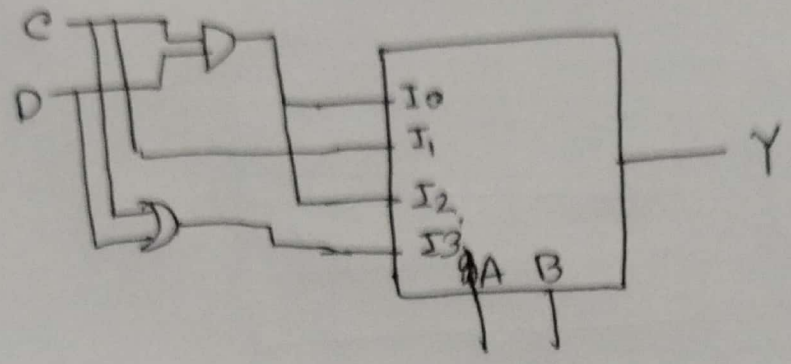
$CD + C\bar{D} + \bar{C}D$

$= C + \bar{C}D$

$= C + CD + \bar{C}D$

$= C + D$

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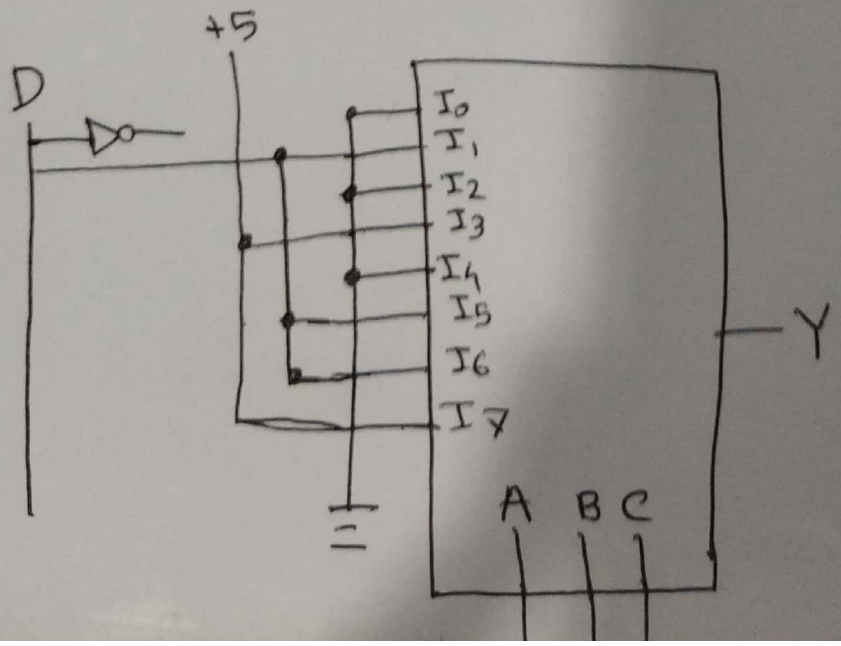


Using 8X1 MUX

Select line → A, B, C

output line → D

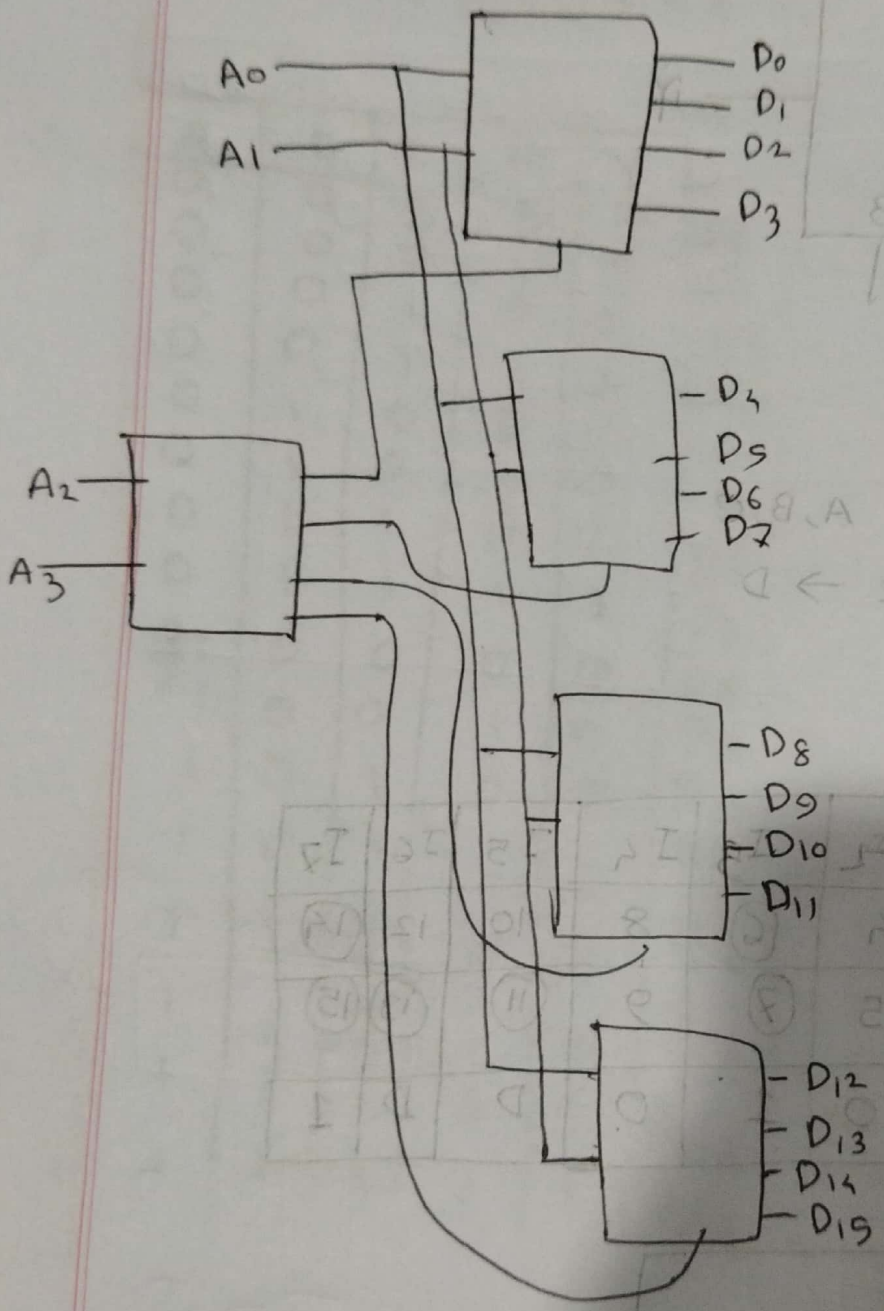
	I_0	I_1	I_2	I_3	I_4	I_5	I_6	I_7
$\bar{D}$	0	2	4	(6)	8	10	12	(14)
D	1	(3)	5	(7)	9	(11)	(13)	(15)
	0	D	0	1	0	D	D	1



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7) a) 4:16 decoder using 2:4 and 3:8

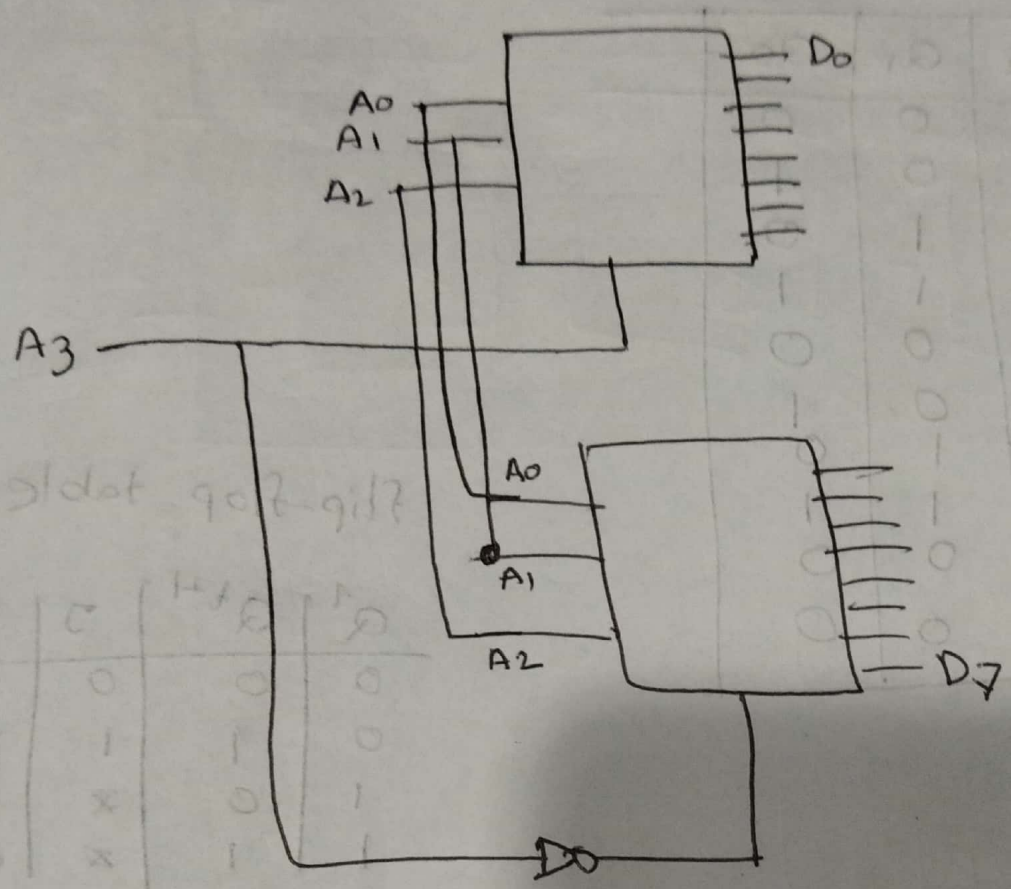
2:4 →



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3 : 8

Input $A_0 A_1 A_2 A_3$



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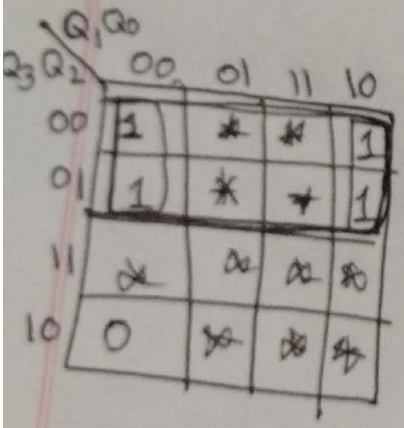
b) MOD-9

clk	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	0	0	0	0

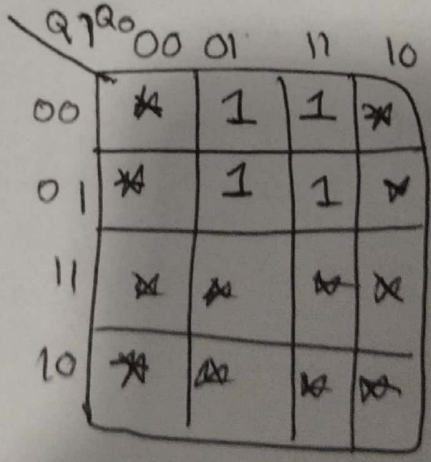
flip-flop table

Q^t	Q^{t+1}	J	k
0	0	0	*
0	1	1	*
1	0	*	1
1	1	*	0

for J_0, k_0



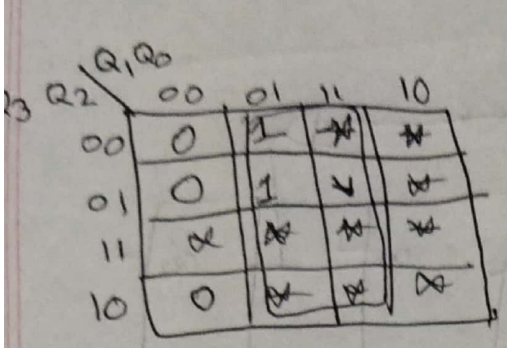
$J_0 = \overline{Q_3}$



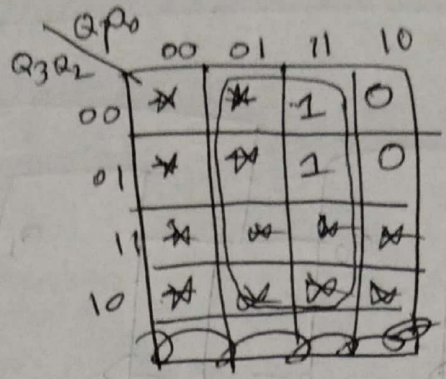
$k_0 = 1$

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$J_1, K_1 \rightarrow$

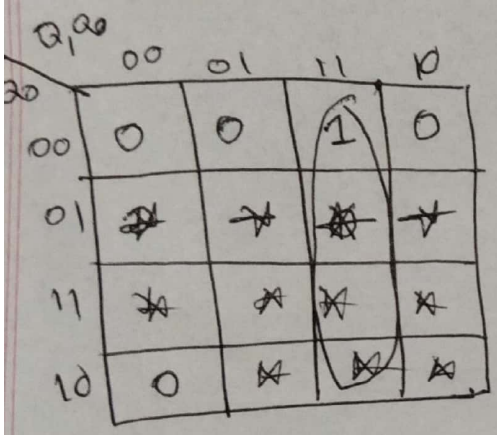


$J_1 = Q_0$

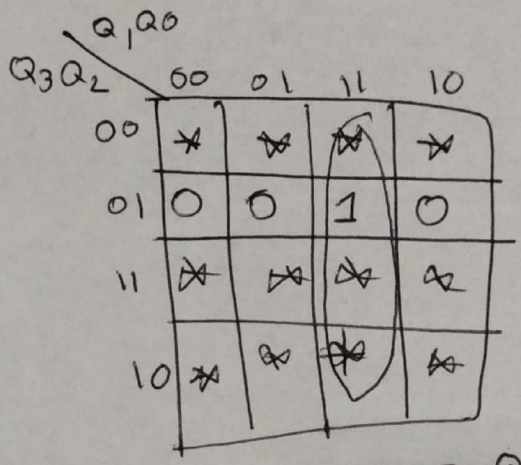


$K_1 = Q_0$

$J_2, K_2 \rightarrow$

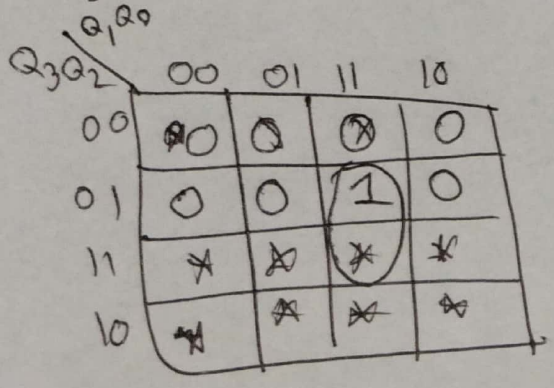


$J_2 = Q_1 Q_0$

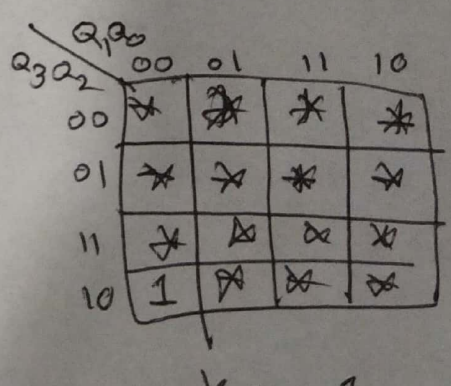


$K_2 = Q_1 Q_0$

J_3, K_3

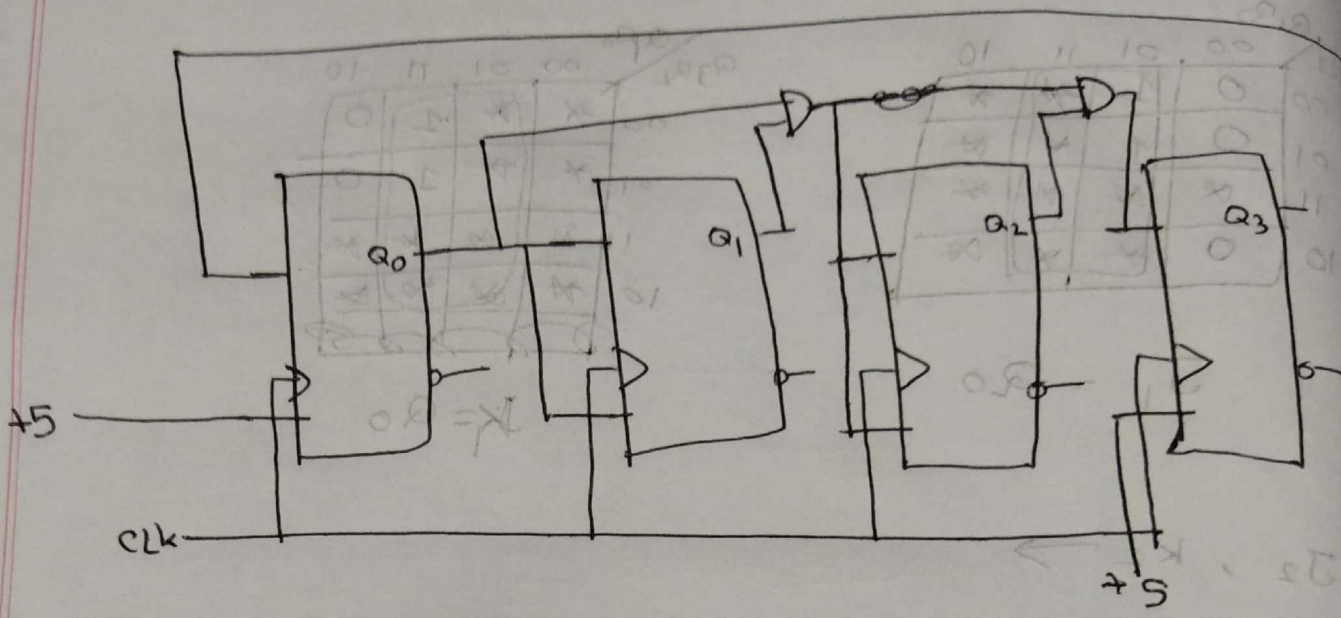


$J_3 = Q_2 Q_1 Q_0$



$K_3 = 1$

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Scanned with CamScanner

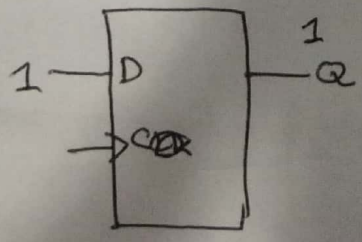
8) a) # Define register with block diagram? Show the basic data movement in shift register?

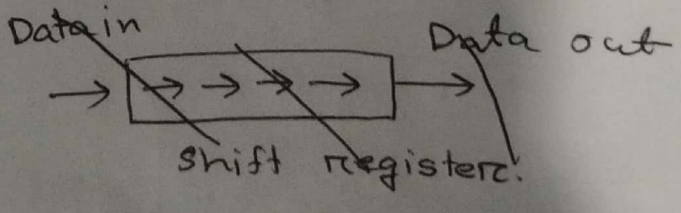
Flip flop is a 1 bit memory cell which can be used for storing the digital data. To increase the storage capacity in terms of number of bits, we have to use a group of flip-flop. Such a group of flip-flop is known as register.

A register has two basic function.

Data storage and data movement.

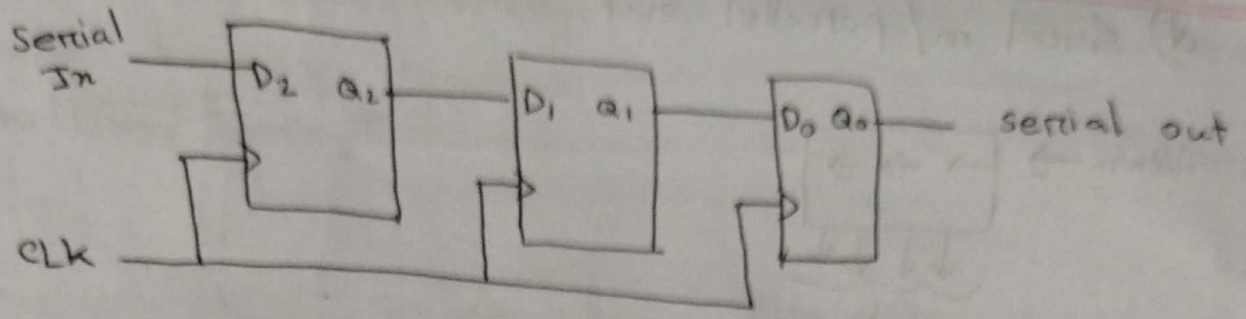
Register can move data from one flip-flop to another. These transfers are shift registers.





1 is stored and appears on output

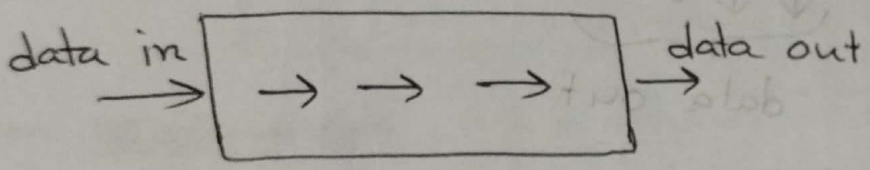
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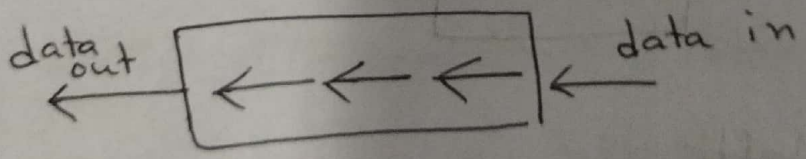
data transferring in register.

Basic data movement in shift register -

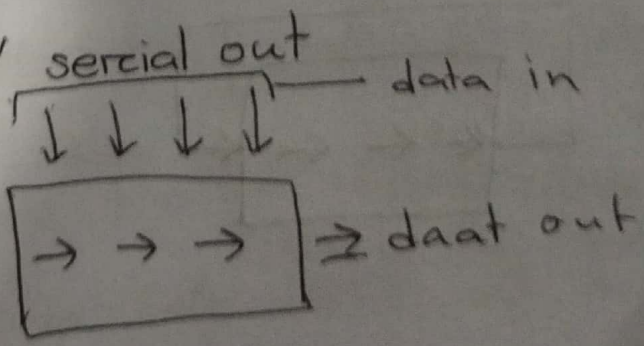
a) Serial in/ shift right/ serial out



b) Serial in/ shift left/ serial out

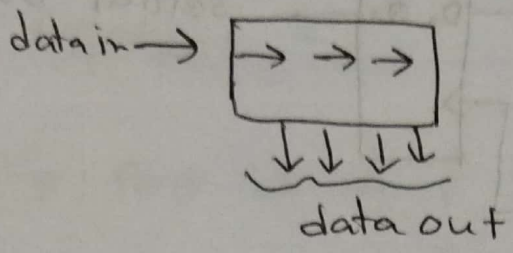


c) Parallel in/ serial out

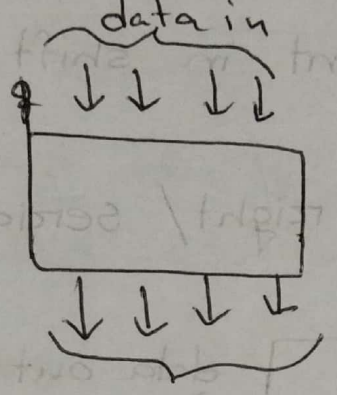


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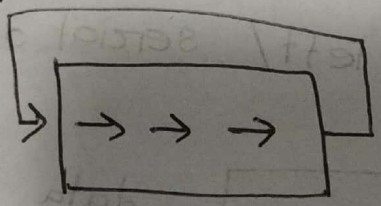
d) Serial in/ parallel out



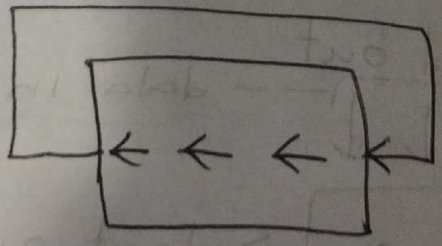
e) parallel in/ parallel out



f) rotate right



g) rotate left

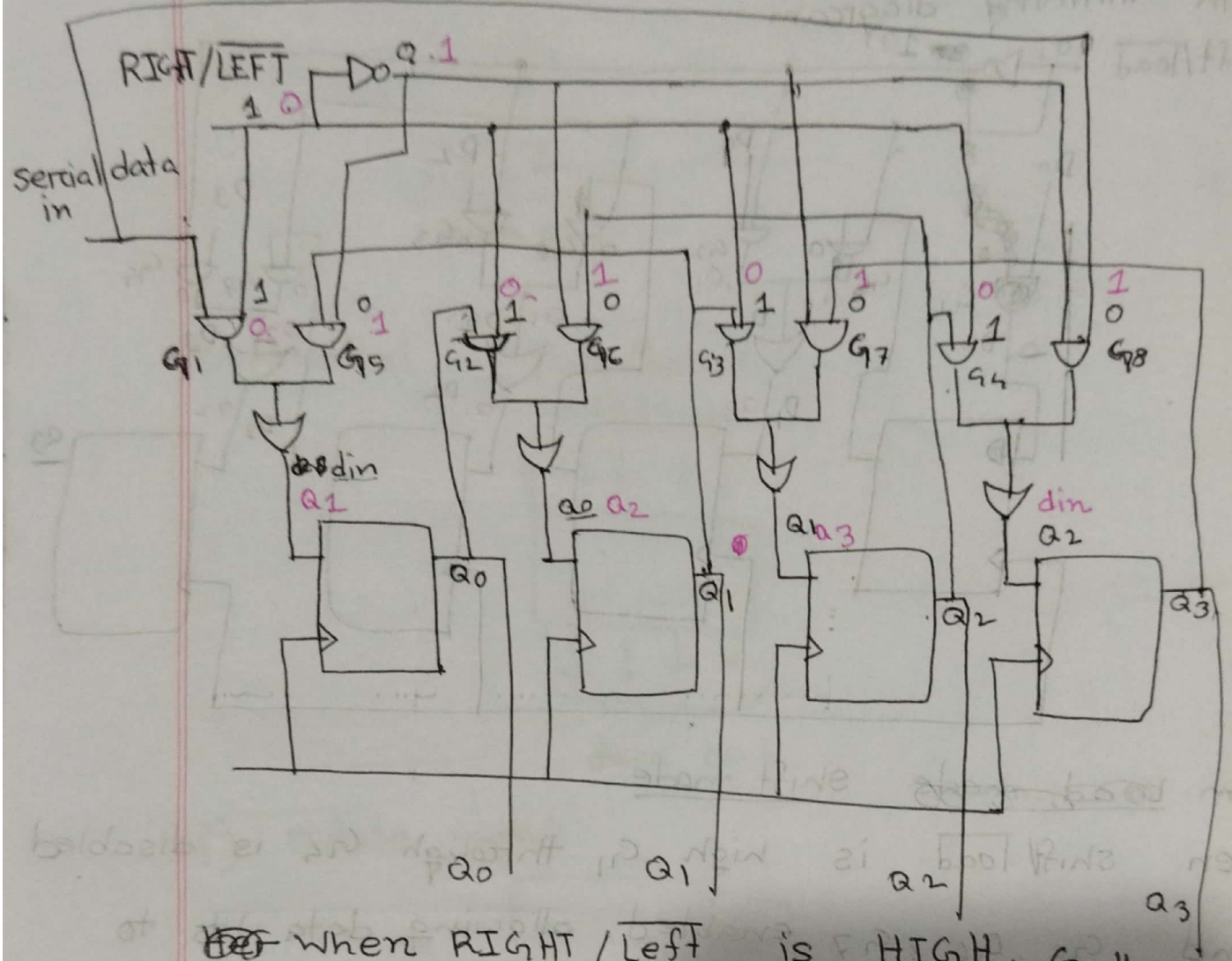


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R1 → din → Q0 → Q1 → Q2 → Q3

L1 → din ← Q3 ← Q2 ← Q1 ← Q0

4 bit bidirectional shift register



When RIGHT / $\overline{Left}$ is HIGH, G_1 through G_4 is enabled and state of Q output for each flip flop is passed through D input of the following flip-flop. when clock pulse occurs, data bits are shifted one place to right.

Scanned with CamScanner

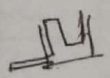
When RIGHT/$\overline{\text{LEFT}}$ is low, G_5 through G_8 is enabled, allowing Q output of each flipflop passing through D input of the following flipflop. when clock pulse occurs the data are shifted one place to left.

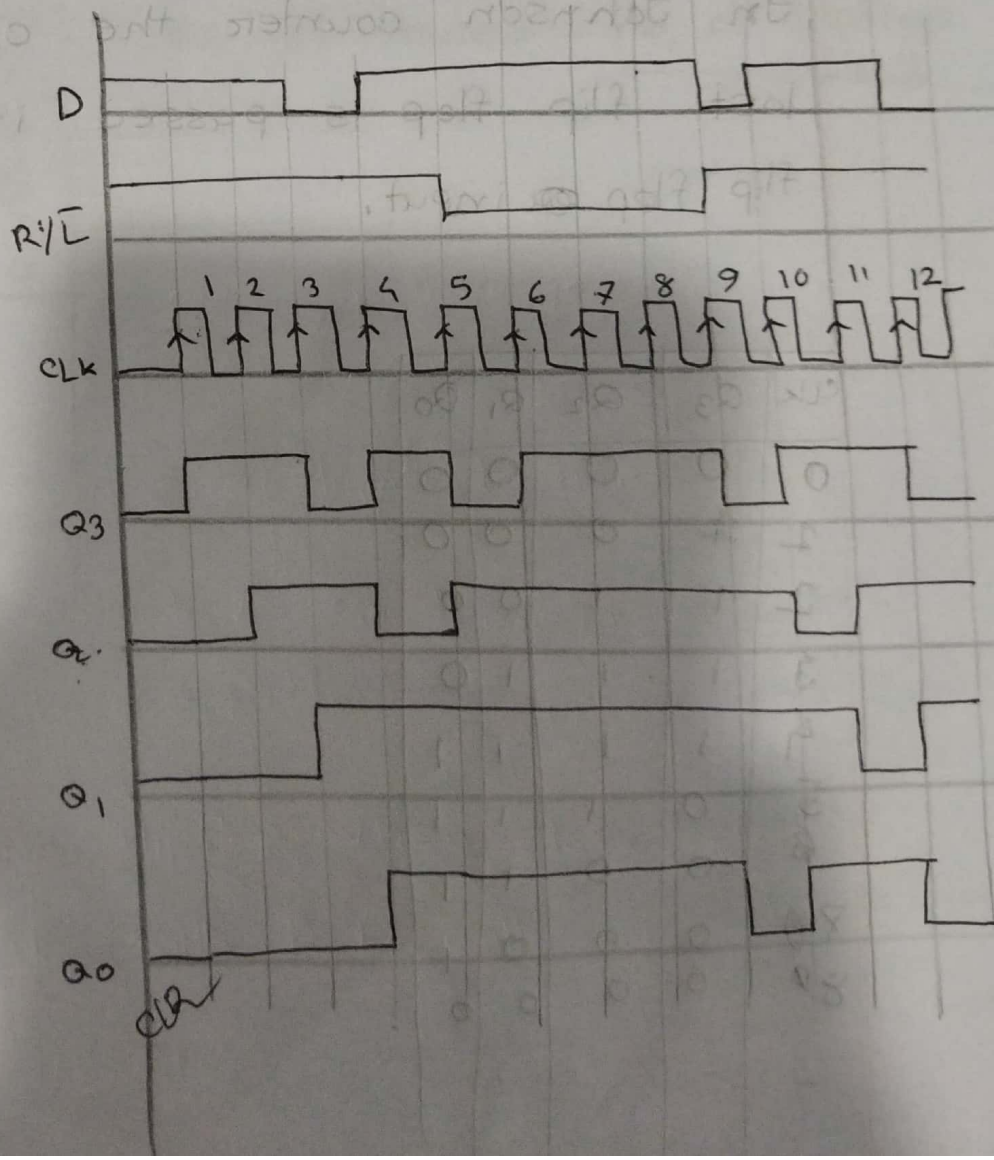
timming diagram

Initial → CLEAR

0 0 0 0

CLK	Q_3	Q_2	Q_1	Q_0	D	
1	1	0	0	0	1	R
2	1	1	0	0	1	R
3	0	1	1	0	0	R
4	1	0	1	1	1	R
5	0	1	0	1	1	L
6	1	1	1	1	1	L
7	1	1	1	1	1	L
8	1	1	1	1	1	L
9	0	1	1	0	0	R
10	1	0	1	1	1	R
11	1	1	0	1	1	R
12	0	1	1	0	0	R





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